REQUIREMENT SPECIFICATION DOCUMENT

IntelliCare: An IoT-based system for household emergency situations

Project Title	IntelliCare: An IoT-based system for household emergency situations
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1. Introduction

This document specifies the requirements for IntelliCare, a smart home IoT safety system. It defines what the system must do and how each component should behave.

Requirements are organised in a hierarchy: a High-Level Requirement (HLR) is broken down into Software Requirements (SW-X) and Sub-requirements (SW-X.Y), each followed by an architectural note (ARCH) that describes the expected behaviour in concrete terms.

1.1 Document Conventions

Tag	Meaning
HLR	High-Level Requirement — the top-level system goal
SW-X	Software Requirement — a traceable, testable functional requirement
SW-X.Y	Sub-requirement that refines SW-X into a specific concern
ARCH	Architectural note — describes the expected runtime behaviour

2. High-Level Requirement

HLR
The system shall provide a smart and reliable solution for detecting and responding to various household hazards, with a focus on improving safety and response time in emergency situations.

3. HLR-1 — User Health and Location Monitoring

HLR-1

The system shall continuously monitor the user's health metrics and track their location within the household environment.

3.1 SW-1.1 — Vital Signs Monitoring

SW-1.1

The system shall utilise sensors integrated into a smart wristband capable of measuring heart rate, blood pressure, and SpO_2 levels, ensuring real-time health monitoring.

3.1.1 SW-1.1.1 — Sensor Hardware

SW-1.1.1

The wristband shall integrate at minimum: an optical heart rate sensor, a blood pressure sensor, and a pulse oximeter for SpO_2 measurement.

ARCH

- The wristband is an embedded device hosting the three biomedical sensors.
- Each sensor is read at a fixed interval (every 5 seconds).
- A single combined health reading is produced per cycle, containing heart rate (bpm), SpO_2 (%), systolic pressure (mmHg) and diastolic pressure (mmHg).

3.1.2 SW-1.1.2 — Data Transmission

SW-1.1.2

The wristband shall transmit health readings to the hub in real time over a wireless connection.

ARCH

- The wristband connects to the home WiFi network and publishes each health reading as an MQTT message.
- Topic: sensors/wristband/health
- QoS 1 (at-least-once delivery); no message retention.
- Each message identifies the sending device, event type, room location, timestamp, and the health data payload.

3.2 SW-1.2 — Motion and Activity Detection

SW-1.2

The system shall detect user movement and activity through the use of an accelerometer and gyroscope embedded within the wristband.

3.2.1 SW-1.2.1 — Motion Sensor Hardware

SW-1.2.1

The wristband shall integrate a 3-axis accelerometer and a 3-axis gyroscope for continuous motion capture.

ARCH

- Both sensors are sampled at 10 Hz (every 100 ms) to capture rapid movements reliably.
- For each sample the device calculates the total acceleration magnitude from all three axes.
- If the magnitude exceeds 2.5 g, the reading is flagged as a potential fall on the device side.

3.2.2 SW-1.2.2 — Motion Data Transmission

SW-1.2.2

The wristband shall transmit motion readings to the hub continuously, including a pre-computed fall-detection flag.

ARCH

- Topic: sensors/wristband/motion
- QoS 1; no retention.
- Each message includes the raw 3-axis values, the computed magnitude, and the fall flag.
- The hub performs the authoritative fall confirmation (see SW-3.1.1).

3.3 SW-1.3 — Indoor Location Tracking

SW-1.3

The system shall determine the user's location within the house using passive infrared (PIR) sensors to enable accurate indoor positioning.

3.3.1 SW-1.3.1 — PIR Sensor Placement

SW-1.3.1

Each room shall contain at least one PIR sensor node capable of detecting occupant presence.

ARCH

- Covered rooms: Kitchen, Living Room, Bedroom, Bathroom, Hallway.
- Each PIR node is a standalone WiFi-connected device.
- Nodes register themselves with the hub automatically at startup using the SDDP discovery protocol.

3.3.2 SW-1.3.2 — Presence Event Publishing

SW-1.3.2

A PIR node shall publish a presence event to the hub whenever motion is detected in its room.

ARCH

- Publishing is event-driven (interrupt on motion detection), not periodic.
- Topic: sensors/{roomID}/pir
- QoS 1; no retention.
- The message includes the room identifier and a boolean motion flag.

3.3.3 SW-1.3.3 — Location Inference at Hub

SW-1.3.3
The hub shall maintain an up-to-date record of the user's current room based on incoming PIR events.

ARCH

- The hub keeps a location map that stores the most recently visited room.
- On each PIR event the hub updates the location entry for the user.
- The inferred location is used by all emergency response workflows.

3.4 SW-1.4 — Automatic Device Discovery

SW-1.4
The system shall utilize the Simple Service Discovery Protocol (SSDP) for automatic identification and registration of all network nodes (wristband, room nodes, and actuators) to the hub.

3.4.1 SW-1.4.1 — SSDP Announcement

SW-1.4.1
Upon power-on, each device shall broadcast an SSDP NOTIFY message to announce its presence and service type to the hub.

3.4.2 SW-1.4.2 — Connectivity Health Check

SW-1.4.2
The hub shall periodically monitor the "alive" status of all registered devices. If a device fails to refresh its presence within the defined timeout, the hub shall mark the device as offline and log a connectivity fault.

4. HLR-2 — Fire Hazard Detection

HLR-2

The system shall be capable of detecting potential fire hazards within the household.

4.1 SW-2.1 — Smoke Detection

SW-2.1

The system shall include an integrated smoke detection sensor in each room, enabling early identification of fire-related risks and supporting timely intervention.

4.1.1 SW-2.1.1 — Smoke Sensor Hardware

SW-2.1.1

Each room node shall embed a smoke sensor capable of measuring smoke concentration in parts per million (ppm).

ARCH

- A photoionisation smoke sensor is used per room.
- The sensor is sampled every 10 seconds under normal conditions.
- If the reading exceeds 50 ppm, the node publishes an alert immediately (interrupt-driven), independently of the regular schedule.

4.1.2 SW-2.1.2 — Smoke Data Transmission

SW-2.1.2

Each room node shall publish its smoke reading to the hub on a shared fire-sensor topic.

ARCH

- Topic: sensors/{roomID}/fire
- QoS 2 (exactly-once) to guarantee delivery of critical readings.
- The payload includes smoke level, CO level, and temperature in a single combined message (all three sensors share one topic and one message per room).

4.2 SW-2.2 — Carbon Monoxide Detection

SW-2.2

The system shall incorporate a carbon monoxide (CO) sensor in every room to continuously monitor air quality and detect the presence of hazardous gases.

4.2.1 SW-2.2.1 — CO Sensor Hardware

SW-2.2.1

Each room node shall embed an electrochemical CO sensor measuring concentration in ppm.

ARCH

- The CO sensor is co-located with the smoke sensor on the same room node.
- Sampled every 10 seconds; immediate publish if reading exceeds 35 ppm.
- The CO value is included in the same combined fire-sensor message (same topic as SW-2.1.2).

4.2.2 SW-2.2.2 — CO Threshold Evaluation

SW-2.2.2

The hub shall evaluate CO readings against a configurable safety threshold and contribute the result to the fire correlation logic.

ARCH

- The hub checks each incoming CO value against CO_WARN_THRESHOLD_PPM (default 35 ppm).
- An exceeded threshold contributes one vote toward the two-of-three fire confirmation rule (see SW-3.2.1).

4.3 SW-2.3 — Temperature Monitoring

SW-2.3

The system shall be equipped with a temperature sensor in each room, allowing continuous monitoring of environmental conditions and early detection of abnormal temperature changes.

4.3.1 SW-2.3.1 — Temperature Sensor Hardware

SW-2.3.1

Each room node shall embed a temperature sensor measuring ambient temperature in degrees Celsius.

ARCH

- A thermistor is co-located with the smoke and CO sensors on the same room node.
- Sampled every 10 seconds; immediate publish if reading exceeds 50 °C.
- The temperature value is included in the combined fire-sensor message.

4.3.2 SW-2.3.2 — Temperature Threshold Evaluation

SW-2.3.2

The hub shall evaluate temperature readings against a configurable threshold and contribute the result to the fire correlation logic.

ARCH

- The hub checks each temperature value against TEMP_WARN_THRESHOLD_C (default 50 °C).
- An exceeded threshold contributes one vote toward the two-of-three fire confirmation rule.

5. HLR-3 — Hazard Detection and Automated Response

HLR-3

The system shall intelligently detect hazardous situations and initiate appropriate responses in real time.

5.1 SW-3.1 — Medical Emergency Detection and Response

SW-3.1

The system shall detect incidents such as falls or abnormal health conditions by analysing data from the wristband sensors. Based on the severity of the situation, the system shall either notify a caregiver or automatically contact emergency medical services. The system shall grant access by unlocking the door.

5.1.1 SW-3.1.1 — Fall Detection

SW-3.1.1

The hub shall confirm a fall event when the wristband signals a high-acceleration impact followed by a period of stillness.

ARCH

- The hub receives motion messages from sensors/wristband/motion.
- When the fall flag in a message is true, the hub starts a 2-second stillness timer.
- If no further motion above 0.3 g is received within those 2 seconds, the fall is confirmed.
- The confirmed fall event carries the user's current room from the location map.

5.1.2 SW-3.1.2 — Health Anomaly Detection

SW-3.1.2

The hub shall detect abnormal vital signs by comparing each incoming health reading against configured thresholds.

ARCH

- The hub evaluates every message on sensors/wristband/health.
- Thresholds:
  - Heart rate below 40 bpm or above 150 bpm → critical alert.
  - SpO_2 below 90 % → critical alert.
  - Systolic blood pressure below 80 mmHg or above 180 mmHg → caregiver alert.
- A critical alert triggers the DIAL_AMBULANCE response; a caregiver alert triggers NOTIFY_FAMILY.

5.1.3 SW-3.1.3 — Caregiver and Emergency Notification

SW-3.1.3

The system shall notify the appropriate party (caregiver or emergency services) depending on the severity of the detected medical event.

ARCH

- The hub publishes a command to the Dialer on topic actuators/dialer.
- For critical events (fall confirmed, SpO_2 critical, heart rate critical): the command instructs the Dialer to call emergency medical services, including the user's room location in the message.
- For caregiver-level events (blood pressure anomaly): the command instructs the Dialer to send an SMS or call the pre-configured caregiver contact.

5.1.4 SW-3.1.4 — Smart Door Access for Emergency Responders

SW-3.1.4

The system shall automatically unlock the front door for emergency personnel.

ARCH

- The hub publishes an UNLOCK command on topic actuators/lock.
- The Lock component receives the command and releases the electromagnetic lock.

5.2 SW-3.2 — Fire Emergency Detection and Response

SW-3.2

In the event of a fire hazard, the system shall activate appropriate safety mechanisms, including sprinkler systems and alarm signals, while simultaneously notifying firefighting services. Additionally, the system shall assist responders by guiding them to the exact location of individuals inside the house and providing controlled access through the smart door system.

5.2.1 SW-3.2.1 — Fire Correlation Logic

SW-3.2.1

The hub shall confirm a fire alert only when at least two of the three environmental sensors (smoke, CO, temperature) in the same room simultaneously exceed their respective thresholds.

ARCH

- For each incoming fire-sensor message the hub evaluates three conditions independently:
  - Smoke level > 50 ppm
  - CO level > 35 ppm
  - Temperature > 50 °C
- Level 1 — WARNING: exactly one condition is true. The hub logs the event and updates the dashboard.
- Level 2 — FIRE: two or more conditions are simultaneously true. The hub triggers the full emergency response for that room.

5.2.2 SW-3.2.2 — Sprinkler Activation

SW-3.2.2

On a confirmed fire alert the system shall activate the sprinkler in the affected room.

ARCH

- The hub publishes a START command to topic actuators/{roomID}/sprinkler.
- The command includes a default activation duration of 60 seconds, after which the sprinkler deactivates unless a further fire alert is received.
- QoS 2 is used to guarantee delivery.

5.2.3 SW-3.2.3 — Alarm Signal Activation

SW-3.2.3

On a confirmed fire alert the system shall activate audible alarms in all rooms.

ARCH

- The hub publishes a TURN_ON command to actuators/{roomID}/speakers with alert type FIRE.
- Each Speaker actuator plays a distinct fire-alarm audio pattern distinguishable from medical alarms.

5.2.4 SW-3.2.4 — Fire Brigade Notification

SW-3.2.4

On a confirmed fire alert the system shall automatically contact the fire brigade, providing the location of the fire.

ARCH

- The hub publishes a DIAL_FIRE_BRIGADE command to actuators/dialer.
- The command payload includes the affected room identifier.
- The Dialer initiates a VoIP or GSM call to emergency services with a pre-recorded or synthesised message stating the address and affected room.

5.2.5 SW-3.2.5 — Responder Guidance via Smart Lighting

SW-3.2.5

The system shall guide emergency responders to the location of the occupant by sequentially activating lights in certain rooms according to user’s last known position.

ARCH

- The hub retrieves the user's current room.
- For that room the hub publishes a TURN_ON / blinking-light command to actuators/{roomID}/lights in sequence.
- The Qt dashboard simultaneously displays the user's room on the house map for additional guidance.

5.2.6 SW-3.2.6 — Smart Door Access for Fire Brigade

SW-3.2.6

On a confirmed fire alert the system shall automatically unlock the front door to allow firefighters access.

ARCH

- The hub publishes an UNLOCK command to actuators/lock with reason fire_brigade.

6. HLR-4 — Centralised Monitoring Dashboard

HLR-4

The system shall provide a centralised dashboard for monitoring all relevant data and system activity.

6.1 SW-4.1 — Real-Time Data Visualisation

SW-4.1

The dashboard shall display real-time health metrics, historical data visualisations, and sensor readings, along with an assessment of whether the user is in a normal or critical condition.

6.1.1 SW-4.1.1 — Health Metrics Display

SW-4.1.1

The dashboard shall display current and historical values for all wristband health metrics.

ARCH

- The Qt dashboard subscribes to dashboard/state.
- Heart rate, SpO_2 , and blood pressure are rendered as live time-series graphs showing the last 60 minutes.
- Each metric has a colour indicator: green for normal, yellow for warning, red for critical.

6.1.2 SW-4.1.2 — Environmental Sensor Display

SW-4.1.2

The dashboard shall display the current smoke, CO, and temperature readings per room.

ARCH

- Each room is shown with a status badge: NORMAL, WARNING, or FIRE.
- The badge colour matches the alert level (green / yellow / red).
- Raw sensor values (ppm, °C) are shown on hover or in a detail panel.

6.1.3 SW-4.1.3 — User Location Display

SW-4.1.3

The dashboard shall display the user's current inferred room on a house map.

ARCH

- A schematic floor plan is rendered with room outlines.
- The user's icon is placed in the room most recently reported active by PIR sensors.
- During a fire emergency the map also highlights the affected room and the responder path.

6.1.4 SW-4.1.4 — System State Overview

SW-4.1.4

The dashboard shall display the real-time connectivity status (Online/Offline) of all sensors and actuators discovered via the SSDP protocol.

ARCH

- The Hub maintains a registry of all active devices discovered via SSDP.
- A dedicated connectivity panel displays a list of all system nodes (wristband, room nodes, actuators) with a real-time status indicator: ONLINE or OFFLINE.
- The Hub marks a device as OFFLINE if its SSDP max-age expires or it fails to respond to an M-SEARCH query within the DEVICE_TIMEOUT_S window.
- Connectivity status changes are published immediately to the dashboard/state topic to ensure the UI reflects the current network health.

6.1.5 SW-4.1.5 — Network Status Display

SW-4.1.5

The dashboard shall show the overall system alert state and the status of key actuators.

ARCH

- A global banner indicates the current alert state: NORMAL, MEDICAL ALERT, or FIRE ALERT.
- Actuator status panels show: sprinkler ON/OFF per room, speaker alarm ON/OFF per room, door lock state (locked / unlocked + reason).
- The hub publishes a full aggregated state message to dashboard/state every 2 seconds and immediately upon any alert state change.

6.2 SW-4.2 — Event Logging

SW-4.2

The system shall maintain a comprehensive log of all detected events, executed actions, and messages received from sensors, ensuring full traceability and system transparency.

6.2.1 SW-4.2.1 — Hub-Side Persistence

SW-4.2.1

The hub shall persist every incoming sensor message and every outgoing actuator command to durable local storage.

ARCH

- Each log entry includes: event category (SENSOR_READING, ALERT_TRIGGERED, ACTUATOR_CMD, DOOR_EVENT), MQTT topic, full message payload, reception timestamp, and a unique correlation ID.
- Storage is append-only to prevent accidental modification of the audit trail.
- New log entries are also streamed to dashboard/logs for live display.

6.2.2 SW-4.2.2 — Dashboard Log Panel

SW-4.2.2
The dashboard shall display the event log in a scrollable, filterable panel.

ARCH

- Entries are displayed in reverse chronological order (newest first).
- Filters available: event category, room, time range, alert severity.
- Each entry shows the timestamp, topic, category badge, and a human-readable summary of the payload.

7. Complete MQTT Topic Map

Topic	Publisher	Subscriber	QoS / Retain
Wristband Sensors			
sensors/wristband/health	Wristband	Hub	QoS 1 / No
sensors/wristband/motion	Wristband	Hub	QoS 1 / No
Room Sensors (wildcard + = roomID)			
sensors/{roomID}/fire	Room node	Hub	QoS 2 / No
sensors/{roomID}/pir	Room node	Hub	QoS 1 / No
Room Actuators			
actuators/{roomID}/sprinkler	Hub	Sprinkler	QoS 2 / No
actuators/{roomID}/speakers	Hub	Speakers	QoS 1 / No
actuators/{roomID}/lights	Hub	Lights	QoS 1 / No
Global Actuators			
actuators/dialer	Hub	Dialer	QoS 2 / No
actuators/lock	Hub	Lock	QoS 2 / No
Dashboard			
dashboard/state	Hub	Dashboard	QoS 1 / Yes
dashboard/logs	Hub	Dashboard	QoS 1 / No

8. Configurable System Parameters

Parameter	Default	Description
HEALTH_SAMPLE_INTERVAL_S	5 s	Wristband vital signs sampling interval
MOTION_SAMPLE_RATE_HZ	10 Hz	Accelerometer / gyroscope sampling frequency
FIRE_SAMPLE_INTERVAL_S	10 s	Room sensor normal polling interval
FALL_ACCEL_THRESHOLD_G	2.5 g	Magnitude threshold for fall pre-detection
FALL_STILLNESS_WINDOW_S	2 s	Post-impact stillness window for fall confirmation
HR_MIN / HR_MAX	40 / 150 bpm	Heart-rate bounds for critical alert
SPO2_CRITICAL_THRESHOLD	90 %	SpO_2 level triggering ambulance call
BP_SYS_MIN / BP_SYS_MAX	80 / 180 mmHg	Systolic BP bounds for caregiver alert
SMOKE_WARN_THRESHOLD_PPM	50 ppm	Smoke level for Level-1 warning
CO_WARN_THRESHOLD_PPM	35 ppm	CO level for Level-1 warning
TEMP_WARN_THRESHOLD_C	50 °C	Temperature for Level-1 warning
SPRINKLER_DEFAULT_DURATION_S	60 s	Default sprinkler activation duration
DASHBOARD_PUBLISH_INTERVAL_S	2 s	Hub to Dashboard periodic state publish interval
SSDP_ALIVE_INTERVAL_S	60 s	Frequency at which devices broadcast their "alive" status (NOTIFY)
SSDP_MAX_AGE_S	150 s	Maximum duration before a device is marked offline if no update is received
HUB_M_SEARCH_INTERVAL_S	300 s	Interval for the hub to issue an active search for new devices
DEVICE_TIMEOUT_S	30 s	Response window for M-SEARCH queries before marking a device offline